

Test Case ID	Feature	Input	Expected Output	Actual Output	Result	Evidence / Link
TC-LA-01	Login & Authentication Valid login (Admin)	Username: osama25, password: osama123	API returns 200 OK with JWT; App navigates to Admin Panel	API returns 200 OK with JWT; App navigates to Admin Panel	Pass	Evidence
TC-LA-02	Login & Authentication Valid login (Manager)	Username: Anas11, password: anas123	API returns 200 OK with JWT; App navigates to Manager Panel	API returns 200 OK with JWT; App navigates to Manager Panel	Pass	Evidence
TC-LA-03	Login & Authentication Valid login (Farmer)	Username: Noor28, password: Noor123	API returns 200 OK with JWT; App navigates to Farmer Panel	API returns 200 OK with JWT; App navigates to Farmer Panel	Pass	evidence
TC-LA-04	Login & Authentication Invalid Credintials	Username: "UnknownUser", Password: "AnyPassword"	API Returns 401Unauthorized ("InvalidUsername or password"), stay at login screen	API Returns 401Unauthorized("InvalidUsername or password"), stay at login screen	Pass	Evidence
TC-LA-05	Login & Authentication Empty Credentials	Username""",password"""	App prevents request or API returns 400 Bad Request; Input validation highlights empty fields.	App prevents request or API returns 400 Bad Request; Input validation highlights empty fields.	pass	evidence
TC-LO-01	Logout Storage & token cleanup	Click Logout button	JWT is purged from Secure Storage and backend marks DeviceToken as inactive/null in the DB to stop notifications.	JWT is purged from Secure Storage and backend marks DeviceToken as inactive/null in the DB to stop notifications.	Pass	evidence
TC-AS-01	Add Sensor Data Valid Data submission	JSON: { Temp: 24, Humidity: 60,SoilMoisture:55, N: 40, P: 35, K: 50, pH: 6.5}	API returns 200; Record visible in DB with current timestamp.	API returns 200; Record visible in DB with current timestamp.	pass	evidence

TC-AS-02	Add Sensor Data Valid Invalid PlantId	JSON: { Temp: 24, Humidity: 60,SoilMoisture:55, N: 40, P: 35, K: 50, pH: 6.5}	API returns 404 Not Found.	API returns 404 Not Found.	Pass	evidence
TC-AS-03	Add Sensor Data (Hardware Integration)	Real-time transmission from ESP32 via Wi-Fi	API successfully parses raw sensor string and logs data to the database	API successfully parses raw sensor string and logs data to the database	Pass	Evidence
TC-AS-04	Add Sensor (Data without API Key)	JSON: { Temp: 24, Humidity: 60,SoilMoisture:55, N: 40, P: 35, K: 50, pH: 6.5}	API Returns 401 Unauthorized	API Returns 401 Unauthorized	Pass	evidence
TC-AP-01	Assigned Plants For Farmer	Log in as Farmer who has assigned plants	App displays only plants assigned to this farmer	App displays only plants assigned to this farmer	Pass	Evidence
TC-AP-02	Assigned Plants For Farmer (No assignments)	Login as a new Farmer with 0 assignments	"No plants assigned"	"No plants assigned"	Pass	evidence
TC-AGP-01	All Greenhouse Plants for Manager	Login as "Manager_X" choose a greenhouse and navigates to all plants	App displays all plants located within Greenhouse_X,	App displays all plants located within Greenhouse_X,	Pass	evidence
TC-AGP-02	All Greenhouse Plants for Manager (No Plants)	Login as "Manager_X" choose a greenhouse and navigates to all plants	App displays "no plants for this greenhouse"	App displays "no plants for this greenhouse"	Pass	evidence
TC-GR-01	Generate Report (PDF)	Date Range: "Last 7 Days", Plants: tomato, Metric: "Soil Moisture"	API generates PDF; File downloads successfully to mobile device storage.	API generates PDF; File downloads successfully to mobile device storage.	Pass	Evidence

TC-GR-02	Generate Report (Excel)	Date Range: "Last 7 Days", Plants: tomato, Metric: "Soil Moisture"	API generates a Excel; File downloads successfully to mobile device storage.	API generates a Excel; File downloads successfully to mobile device storage.	Pass	evidence
TC-GR-03	Generate Report Empty Data Range	Date Range: Future Dates (e.g., 2027)	API Generates a file with no data	API Generates a file with no data	Pass	evidence
TC-ST-01	Sensor Trends	"Date Range:" Last Two Dayes" Metrics:"Tem perature", "Hu midity", "PH"	API Returns Sensor Data and the app displays them in chart	API Returns Sensor Data and the app displays them in chart	Pass	Evidence
TC-ST-02	Sensor Trends (No Data)	"DateRange And Metrics"	API Returns 404 No Sensor Data	API Returns 404 No Sensor Data	Pass	evidence
TC-ST-03	Sensor Trends (Date Range older than two months)	"DateRange And Metrics"	API Returns 400	API Returns 400	Pass	evidence
TC-ST-04	Sensor Trends (Date Range in the future)	"DateRange And Metrics"	API Returns 400	API Returns 400	Pass	evidence
TC-AST-01	Archive Sensor Trends	"DateRange,P lant,Metrics"	API Returns Sensor Data and the app displays them in chart	API Returns Sensor Data and the app displays them in chart	Pass	evidence

TC-AST-02	Archive Sensor Trends (No Data)	“DateRange,Plant,Metrics”	API Returns 404 No Sensor Data	API Returns 404 No Sensor Data	Pass	evidence
TC-AST-03	Archive Sensor Trends (Date Range in the future)	“DateRange,Plant,Metrics”	API Returns 400	API Returns 400	Pass	evidence
TC-AST-04	Archive Sensor Trends (after one year or before two months)	“DateRange,Plant,Metrics”	API Returns 400	API Returns 400	Pass	evidence
TC-MG-01	Manage Greenhouses (Add)	GreenhouseName:Greenhouses_A,Location:Amman,image	API Returns Success and the greenhouse appeared in the list	API Returns Success and the greenhouse appeared in the list	Pass	Evidence
TC-MG-02	Manage Greenhouses (Add)	Greenhouse Name: Greenhouse B, Location: <i>Empty</i>	API returns validation error and greenhouse is not added	API returns validation error and greenhouse is not added	PASS	evidence
TC-MG-03	Manage Greenhouses (Add)	Greenhouse Name: Greenhouse_C, Location: Amman, Image: Not Uploaded	API returns success and the greenhouse is added without image	API returns success and the greenhouse is added without image	Pass	evidence
TC-MG-04	Manage Greenhouses (update)	selects an existing greenhouse and uploads a new image	API returns success and the greenhouse image is updated successfully	API returns success and the greenhouse image is updated successfully	Pass	evidence

TC-MPT-01	Manage plant Type(add)	Plant Name: Eggplant,	API returns success and the plant appears in the plant list	API returns success and the plant appears in the plant list	Pass	evidence
TC-MPT-02	Manage plant Type(update)	selects an existing plant and updates plant details	API returns success and the plant information is updated successfully	API returns success and the plant information is updated successfully	Pass	evidence
TC-MPT-03	Manage Plant Type(delete)	Admin selects an existing plant and clicks Delete	API returns success and the plant is deleted from the system	API returns success and the plant is deleted from the system	Pass	evidence
TC-MPT-04	Manage Plant Type(delete)	Admin tries to delete a plant that is linked to active data	API returns error Cannot delete plant type with associated plant	API returns error Cannot delete plant type with associated plant	Pass	evidence
TC-MP-01	Manage Plants (Add)	“plantName:tomato3,Plant Type:tomato, Location:low side, image”	API Returns 200 and display all the plants in the greenhouse	API Returns 200 and display all the plants in the greenhouse	Pass	evidence
TC-MP-02	Manage Plants (Update)	“plantName:tomato4,remove image”	API Returns 200 and display all the plants in the greenhouse	API Returns 200 and display all the plants in the greenhouse	Pass	evidence
TC-MP-03	ManagePlants(Delete)	Delete Plant	API Returns 200 and display all the plants in the greenhouse	API Returns 200 and display all the plants in the greenhouse	Pass	evidence

TC-MP-04	ManagePlants(Delete with assigned farmers)	Delete Plant	API Returns 400 and display all the plants in the greenhouse	API Returns 400 and display all the plants in the greenhouse	Pass	evidence
TC-MM-01	Manage Managers (add)	Manager Name: Nasser Hani Username: nhani, Password: Ahmed@123, Role: Manager	API returns success and the manager account is created successfully	API returns success and the manager account is created successfully	Pass	evidence
TC-MM-02	Manage Managers (add with assign green house)	Manager: Ahmed Ali, Greenhouse: Greenhouse_A	API returns success and the manager is assigned to the selected greenhouse	API returns success and the manager is assigned to the selected greenhouse	Pass	evidence
TC-MM-03	Manage Manager (delete)	Admin selects an existing manager account and clicks Delete	API returns success and the manager account is deleted from the system	API returns success and the manager account is deleted from the system	Pass	evidence
TC-MM-04	Manage Manager (delete)	Admin tries to delete a manager account that is linked to active data / does not exist	API returns error and the manager account is not deleted	API returns error and the manager account is not deleted	Pass	evidence
TC-AM-01	Manager Assignment on Greenhouse(Assign Manager)	Pick Manage to assign for the greenhouse	API Returns 200 and display the manage info	API Returns 200 and display the manage info	Pass	evidence
TC-AM-02	Manager Assignment on Greenhouse(UnAssign Manager)	Click on Unassign manager	API Returns 200 and display the mangers to pick	API Returns 200 and display the mangers to pick	Pass	evidence

TC-MF-01	Manage Farmers(Add)	Enters farmer name,username,password and click add	API Returns 200 and the app display all the farmers	API Returns 200 and the app display all the farmers	pass	evidence
TC-MF-02	Manage Farmers(Add with assigned plants)	Enters farmer name,username,password and click add	API Returns 200 and the app display all the farmers	API Returns 200 and the app display all the farmers	pass	evidence
TC-MF-03	Manage Farmer (delete)	Manager selects an existing farmer account and clicks Delete	API returns success and the farmer account is deleted from the system	API returns success and the farmer account is deleted from the system	Pass	evidence
TC-MF-04	Manage Farmer (delete)	Manager tries to delete a farmer account that is linked to active data	API returns error and the farmer account is not deleted	API returns error and the farmer account is not deleted	Pass	evidence
TC-MPA-01	Manage Plant Assignment (Assign)	Manager selects one or many farmers to assign them on plant	API return 200 and display the plant with assigned farmer	API return 200 and display the plant with assigned farmer	Pass	evidence
TC-MPA-02	Manage Plant Assignment (UnAssign)	Manager click delete on a specific farmer from a plant to unassing	API return 200 and display the plant with assigned farmer	API return 200 and display the plant with assigned farmer	Pass	evidence
TC-MPS-01	Manage Plant Schedules (Add)	Selects task type , time and days	API return 200 and display the plant schedules	API return 200 and display the plant schedules	Pass	evidence

TC-MPS-02	Manage Plant Schedules (Update)	Selects task type , time and days	API return 200 and display the plant schedules	API return 200 and display the plant schedules	Pass	evidence
TC-MPS-03	Manage Plant Schedules (Toggle off)	Click on the toggle icon	API Returns 200	API Returns 200	Pass	evidence
TC-MPS-04	Manage Plant Schedules (Toggle onn)	Click on the toggle icon	API Returns 200	API Returns 200	Pass	evidence
TC-MPS-05	Manage Plant Schedules (Delete)	Click on the delete icon for a schedule	API Returns 200 and displays the schedules	API Returns 200 and displays the schedules	Pass	evidence
TC-CUI-01	Update User Info	User updates profile details (Name / Email / Phone)	API returns success and profile information is updated successfully	API returns success and profile information is updated successfully	pass	evidence
TC-CP-01	Change Password	Current Password: User@123, New Password: User@456, Confirm Password: User@456	API returns success and password is updated successfully	API returns success and password is updated successfully	Pass	
TC-CP-02	Change Password	Current Password: User@123, New Password: User@123,	API returns validation error and password is not changed	API returns validation error and password is not changed	Pass	evidence

		Confirm Password: User@123				
TC-CP-03	Change Password	Current Password: User@123, New Password: 123, Confirm Password: 123	API returns validation error and password is not changed	API returns validation error and password is not changed	Pass	evidence
TC-SIN-01	Sensor Issue Notification	The sensor data sent to the server but the dht sensor is sending null values	API Returns 200 after adding sensor data but a notification sent to the manager	API Returns 200 after adding sensor data but a notification sent to the manager	Pass	evidence
TC-SIN-02	Sensor Issue Notification	The sensor data sent to the server but the soil sensor is sending null values	API Returns 200 after adding sensor data but a notification sent to the manager	API Returns 200 after adding sensor data but a notification sent to the manager	Pass	evidence
TC-PN-01	Plant Notifications	User Clicks on plant's notification icon	API Returns all plant's notifications and display them	API Returns all plant's notifications and display them	Pass	Evidence
TC-SR-01	Greenhouse System reports for manager	Manager clicks on the system reports	API Reutrns all system report for this greenhouse	API Reutrns all system report for this greenhouse	Pass	evidence
TC-SRA-01	System Reports For Admin	Admin clicks on System Reports	API Returns All Greenhouses System Reports And display them	API Returns All Greenhouses System Reports And display them	Pass	Evidence

